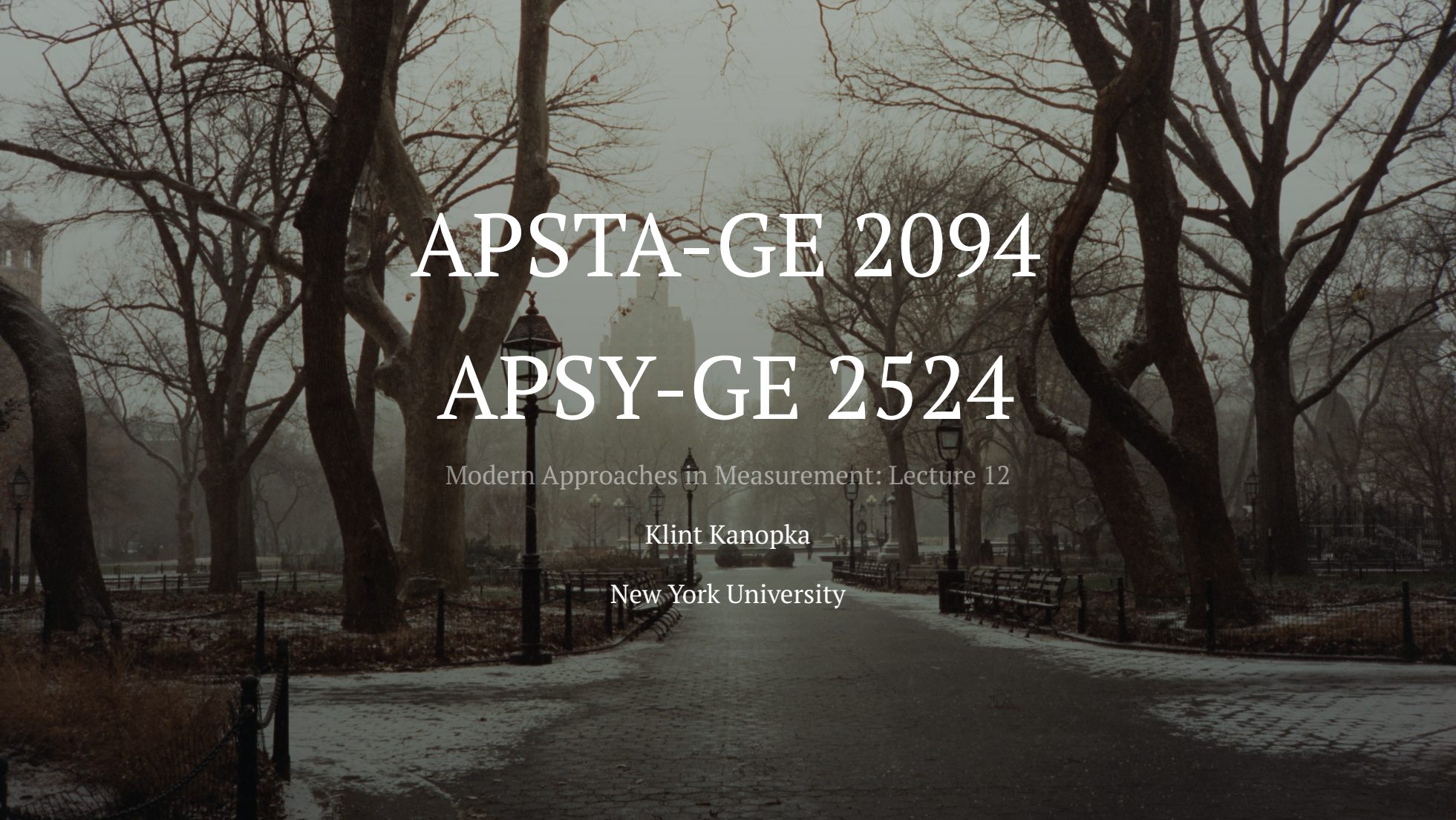




APSTA-GE 2094

APSY-GE 2524

Modern Approaches in Measurement: Lecture 12

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New York University

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- Another note: As we wind down, think about any specific topics or applications you're interested in or might want to learn more about

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 - Stars, thumbs up/down, swiping left/right, etc.

The Utility Matrix

- Ratings are stored in an object called the *utility matrix*

	Taylor Swift	Bad Bunny	Turnstile	Pantera	Crowbar
Alice	5	3	3	1	1
Bob	1	1	1	5	4
Carol	4	4	5	3	2
David	3	2	4	4	5

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- Does this remind you of anything we've dealt with before?

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 - Upside is that you only really care about predicting high unknown ratings!
 - Downside is that this is a tricky problem and measuring success isn't always straightforward
 - Any early ideas on how this can be done?

Movie Recommendation

- With a group of 3 ± 1 :

1. Download the MovieLens 100K Dataset
2. Unzip the files, check out the `README` to get a sense of what's in there and how the data are organized
3. The full dataset is in `u.data`. Using whatever you like from the files provided, come up with a plan for recommending new movies to users
4. Try to code up a couple of test cases or mild proof-of-concepts
5. Start with something very simple and then add complexity!

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- How do you decide on similarity?
 - Use a distance metric!

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- How might this work in practice?

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 - You can take a weighted average using their similarity to x

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- Which version do you think works better? Why?

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- Which version do you think works better? Why?
 - Item-item usually works better, as items are simpler

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 - Tends to over recommend popular items

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 - Talk to a few neighbors about it!

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- Now they display people based on app usage

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 - This requires group members to develop rankings of members of the other group

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- Guaranteed to produce stable matches

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- The OKCupid Data Blog used to be full of cool stuff, but it's really only accessible via the Wayback Machine

Wrap Up

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- CDMs combine latent class analysis with theory to model categorical latent skill profiles
- In CDMs, class membership is defined by which attributes a respondent possesses
- Process data are collected while respondents interact with items
- Response time is the most common form of process data and can be useful if we have a clear plan for using it